Dysprosium-163

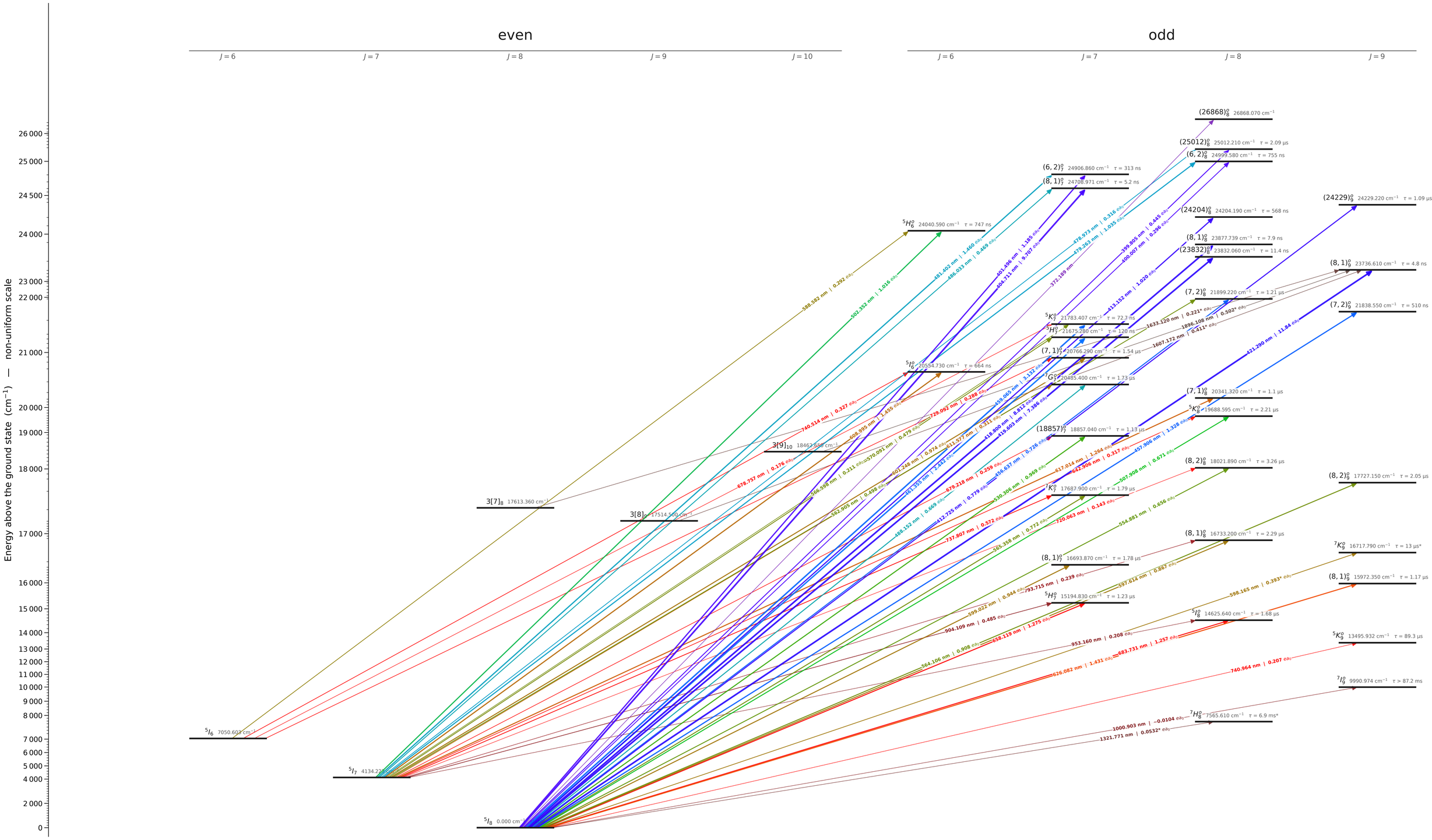
^{163}Dy (Dy I), nuclear spin I = 5/2. Energy levels and transitions below 2 μm: 39 levels, 55 lines.

Dy^{+} ionisation limit 47901.76 cm^{-1} = 5.93906 eV (208.761 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $|\langle J\|er\|J'\rangle|$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

Arrow colour = vacuum wavelength (nm)

300 400 500 600 700 800 1000 1200 1500 2000



Key transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$\|\langle J\|er\|J'\rangle\|$ (ea_0)	A (s^{-1})	Γ/2π up. (MHz)	branch (%)	Use
$^5I_8$ – (26868)$^9_8$	372.1890	372.0832	805.4845	–	–	–	–	UV ground-state line
$^5I_8$ – (8, 1)$^9_7$	404.7113	404.5970	740.7563	9.707	1.920e+08	30.61	99.8	strong blue J=8->7 line
$^5I_8$ – (8, 1)$^9_8$	418.8001	418.6821	715.8366	8.812	1.260e+08	20.15	99.5	strong blue J=8->8 line
$^5I_8$ – (23832)$^9_8$	419.6028	419.4846	714.4672	7.386	8.800e+07	13.96	100	strong blue J=8->8 line
$^5I_8$ – (8, 1)$^9_9$	421.2902	421.1715	711.6057	11.84	2.000e+08	33.16	100	main broad cooling line: Zeeman slower, transverse coolin…
$^5I_8$ – (7, 2)$^9_8$	456.6373	456.5093	656.5221	0.726	6.600e+05	0.1315	79.9	pure 6s->6p reference line for King plots
$^5I_8$ – (8, 2)$^9_9$	564.1065	563.9499	531.4466	0.908	4.900e+05	0.07764	100	closed J=8->9 narrow line
$^5I_8$ – $^7K^o_9$	598.1652	597.9995	501.1867	0.393*	7.700e+04	0.01224	100	"598 nm" closed J=8->9 narrow line proposed for narrow-li…
$^5I_8$ – (8, 1)$^9_7$	599.0223	598.8564	500.4696	0.944	5.600e+05	0.08941	99.7	narrow J=8->7 probe line
$^5I_8$ – (8, 1)$^9_9$	626.0819	625.9088	478.8390	1.431	8.900e+05	0.136	100	narrow-line MOT
$^5I_8$ – $^5H^o_7$	658.1186	657.9368	455.5295	1.275	7.700e+05	0.1294	94.7	narrow probe line used for Doppler thermometry of the 421…
$^5I_8$ – $^5I^o_8$	683.7308	683.5421	438.4657	1.257	5.893e+05	0.09474	99	narrow J=8->8 optical pumping / demagnetization cooling l…
$^5I_8$ – $^5K^o_9$	740.9640	740.7600	404.5979	0.207	1.120e+04	0.001782	100	ultra-narrow-line MOT / narrow-line cooling in tweezers
$^5I_8$ – $^7I^o_9$	1000.9034	1000.6291	299.5219	0.010	1.150e+01	1.825e-06	100	ultra-narrow
$^5I_8$ – $^7H^o_8$	1321.7705	1321.4091	226.8113	0.053*	1.460e+02	2.307e-05	100	proposed ultra-narrow telecom-band line

λ, ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
Γ/2π: natural linewidth of the upper level, 1/(2πτ).

^{163}Dy hyperfine structure (I = 5/2)

Level	A (MHz)	B (MHz)	Shift of each F level from the centre of gravity (MHz)
$^5I_8$	162.7540(20)	1152.87(4)	F=11/2: -3251.26 F=13/2: -2549.30 F=15/2: -1588.04 F=17/2: -302.63 F=19/2: +1380.44 F=21/2: +3543.30
$^7H^o_8$	131.000	1901.000	F=11/2: -2270.27 F=13/2: -2005.70 F=15/2: -1450.93 F=17/2: -499.01 F=19/2: +971.23 F=21/2: +3095.25
$^7I^o_9$	125.000	2901.000	F=13/2: -2129.56 F=15/2: -2087.96 F=17/2: -1654.01 F=19/2: -682.66 F=21/2: +988.20 F=23/2: +3537.75
$^5K^o_9$	142.91(3)	4105.0(20)	F=13/2: -2164.17 F=15/2: -2360.07 F=17/2: -2034.75 F=19/2: -982.97 F=21/2: +1024.68 F=23/2: +4241.73
$^5I^o_8$	152.560(20)	2357.0(10)	F=11/2: -2592.92 F=13/2: -2329.00 F=15/2: -1715.13 F=17/2: -618.71 F=19/2: +1110.50 F=21/2: +3640.45
(8, 1)$^9_9$	261.700	-334.000	F=13/2: -6657.11 F=15/2: -4591.21 F=17/2: -2294.39 F=19/2: +216.64 F=21/2: +2923.23 F=23/2: +5804.75
(8, 1)$^9_7$	89.858(17)	943.2(9)	F=9/2: -1444.76 F=11/2: -1241.27 F=13/2: -879.52 F=15/2: -298.87 F=17/2: +570.65 F=19/2: +1808.32
(8, 2)$^9_8$	207.295(13)	1652.2(9)	F=13/2: -4615.44 F=15/2: -3570.97 F=17/2: -2166.94 F=19/2: -320.74 F=21/2: +2059.95 F=23/2: +5077.19
(8, 1)$^9_8$	121.620(20)	1844.9(4)	F=13/2: -2407.45 F=15/2: -2065.05 F=17/2: -1431.00 F=19/2: -413.08 F=21/2: +1091.83 F=23/2: +3197.68
(26868)$^9_8$	110.3(12)	1595(57)	F=11/2: -1913.53 F=13/2: -1689.04 F=15/2: -1220.66 F=17/2: -418.69 F=19/2: +818.57 F=21/2: +2604.75

Measured A, B constants; sources in the data file.

Isotope shifts

Transition	λ vac (nm)	Isotopes	Shift (MHz)
$^5I_8$ – (26868)$^9_8$	372.189	164-163	-44(10)
$^5I_8$ – (26868)$^9_8$	372.189	164-162	-178(13)
$^5I_8$ – (26868)$^9_8$	372.189	164-161	-194(13)
$^5I_8$ – (26868)$^9_8$	372.189	164-160	-339(13)
$^5I_8$ – (8, 1)$^9_9$	421.290	164-163	-616.3(5)
$^5I_8$ – (8, 1)$^9_9$	421.290	164-162	-913.2(8)
$^5I_8$ – (8, 1)$^9_9$	421.290	164-161	-1635.0(10)
$^5I_8$ – (8, 1)$^9_9$	421.290	164-160	-1895.0(20)
$^5I_8$ – (8, 1)$^9_9$	421.290	164-158	-2868(9)
$^5I_8$ – (8, 1)$^9_9$	421.290	164-156	-4300(15)
$^5I_8$ – (7, 2)$^9_8$	456.637	164-163	-660(3)
$^5I_8$ – (7, 2)$^9_8$	456.637	164-162	-971.0(20)
$^5I_8$ – (7, 2)$^9_8$	456.637	164-161	-1744(3)
$^5I_8$ – (7, 2)$^9_8$	456.637	164-160	-2020(3)
$^5I_8$ – (7, 2)$^9_8$	456.637	164-158	-3061(4)
$^5I_8$ – (7, 2)$^9_8$	456.637	164-156	-4604(5)
$^5I_8$ – (8, 2)$^9_9$	564.106	164-163	-685.0(20)
$^5I_8$ – (8, 2)$^9_9$	564.106	163-162	-328(3)
$^5I_8$ – (8, 2)$^9_9$	564.106	162-161	-801(3)
$^5I_8$ – (8, 2)$^9_9$	564.106	161-160	-284.0(20)
$^5I_8$ – (8, 2)$^9_9$	564.106	164-162	-1012.0(20)
$^5I_8$ – (8, 2)$^9_9$	564.106	162-160	-1086(5)

Shift = ν(first isotope) − ν(second isotope).